Diskrete Mathematik – Woche 7

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https://github.com/tschai-yim/eth-resources
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Functions

Problem 1 ⋆

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$.

1. Which of the following relations from A to B are functions $A \to B$? Which are partial functions (Def. 3.35)?

$$\rho_1 = \{(1, x), (2, y), (3, x)\}, \quad \rho_2 = \{(1, x), (2, y)\}, \quad \rho_3 = \{(1, x), (1, y), (2, y), (3, y)\}, \quad \rho_4 = \varnothing$$

2. Determine $|B^A|$, the number of partial functions $A \to B$, and the number of relations from A to B.
3. Let $C \neq \varnothing$ be a set. Determine the sets $C^\varnothing$, $\varnothing^C$ and $\varnothing^\varnothing$ (Def. 3.34).
4. Which of the following relations on $\mathbb{R}$ are functions $\mathbb{R} \to \mathbb{R}$?

$$\sigma_1 = \{(u, v) \mid v = u^2\}, \quad \sigma_2 = \{(u, v) \mid u = v^2\}, \quad \sigma_3 = \{(u, v) \mid u^2 + v^2 = 1\}$$

Solution:

1. Check Def. 3.33: (1) totally defined, (2) well-defined. Partial function = only (2).
   - ρ_1: function (every $a \in A$ has exactly one partner).
   - ρ_2: partial function, not a function: 3 has no partner, so (1) fails.
   - ρ_3: neither: $1 \rho_3 x$ and $1 \rho_3 y$ with $x \neq y$, so (2) fails.
   - ρ_4: partial function ((2) holds vacuously), not a function: (1) fails for every $a \in A$.
2. - $|B^A| = |B|^{|A|} = 2^3 = 8$: each of the 3 arguments independently gets one of 2 values.
   - Partial functions: each argument gets x, y or "undefined": $3^3 = 27$.
   - Relations: subsets of $A \times B$: $2^{|A \times B|} = 2^6 = 64$.
3. - $C^\varnothing = \{\varnothing\}$. The only relation from $\varnothing$ to C is $\varnothing \subseteq \varnothing \times C = \varnothing$, and both conditions of Def. 3.33 start with $\forall a \in \varnothing$, so they hold vacuously.
   - $\varnothing^C = \varnothing$. The only candidate is again $\varnothing$, but condition (1) requires $\forall a \in C \ \exists b \in \varnothing \ldots$, which is false since $C \neq \varnothing$.
   - $\varnothing^\varnothing = \{\varnothing\}$ (vacuous as in the first case).
   - Consistent with the counting rule $|B^A| = |B|^{|A|}$: $|C|^0 = 1$, $0^{|C|} = 0$, $0^0 = 1$.
4. - σ_1: function ($u \mapsto u^2$).
   - σ_2: not a function. Not totally defined: no v with $v^2 = -1$. Not well-defined: $4 \sigma_2 2$ and $4 \sigma_2 (-2)$.
   - σ_3 (the unit circle, Example 3.12): not a function. 2 has no partner, and $0 \sigma_3 1$, $0 \sigma_3 (-1)$.
   - Note: restricted to $\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$, σ_2 is the function $u \mapsto \sqrt{u}$. Whether a relation is a function depends on domain and codomain.

Problem 2 ⋆ ⋆

Let $f : A \to B$ be a function and $S, S' \subseteq A$. Prove or disprove:

1. $f(S \cup S') = f(S) \cup f(S')$
2. $f(S \cap S') = f(S) \cap f(S')$

3. ($\star\star\star$) Prove: f is injective $\Leftrightarrow f(S \cap S') = f(S) \cap f(S')$ for all $S, S' \subseteq A$.

Solution:

Equalities of sets are shown via Def. 3.2 (or both inclusions, Lemma 3.2).

1. **True.** For $b \in B$:

$$\begin{aligned} b \in f(S \cup S') &\Leftrightarrow \exists a(a \in S \cup S' \wedge f(a) = b) && \text{(Def. 3.36)} \\ &\Leftrightarrow (\exists a \in S : f(a) = b) \vee (\exists a \in S' : f(a) = b) && \text{(cases } a \in S, a \in S') \\ &\Leftrightarrow b \in f(S) \cup f(S') && \text{(Def. 3.36, 3.4)} \end{aligned}$$

2. **False**, only $\subseteq$ holds. Let $A = \{1, 2\}$, $B = \{0\}$, $f(1) = f(2) = 0$, $S = \{1\}$, $S' = \{2\}$:
   - $f(S \cap S') = f(\varnothing) = \varnothing$, but $f(S) \cap f(S') = \{0\} \cap \{0\} = \{0\}$.
   - ($\subseteq$ always holds: $b = f(a)$ with $a \in S \cap S'$ gives $b \in f(S)$ and $b \in f(S')$.)
3. ($\Rightarrow$) $\subseteq$ holds always (see 2.). For $\supseteq$ let $b \in f(S) \cap f(S')$, i.e. $b = f(a) = f(a')$ with $a \in S$, $a' \in S'$.
   - f injective (Def. 3.39, contrapositive form: $f(a) = f(a') \Rightarrow a = a'$) gives $a = a' \in S \cap S'$.
   - Hence $b = f(a) \in f(S \cap S')$.

   ($\Leftarrow$) Contrapositive. If f is not injective, there are $a \neq a'$ with $f(a) = f(a')$.
   - Choose $S = \{a\}$, $S' = \{a'\}$: $f(S \cap S') = f(\varnothing) = \varnothing$, but $f(S) \cap f(S') = \{f(a)\} \neq \varnothing$.

Problem 3 $\star\star$

Let $f : A \to B$ and $g : B \to C$ be functions.

1. Express the function $g \circ f$ (Def. 3.41) as a composition of relations in the sense of Def. 3.12.
2. Prove: if f and g are injective, then $g \circ f$ is injective. If f and g are surjective, then $g \circ f$ is surjective.
   (The first statement is used without proof in Lemma 3.15 (ii).)
3. Prove: if $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.
4. Find A, B, C, f, g such that $g \circ f$ is bijective, but f is not surjective and g is not injective.

Solution:

1. As a set of pairs, the function $g \circ f$ is the relation composition $f \circ g$ of Def. 3.12 (footnote 47).
   - $(a, c) \in f \circ g$ (relations) $\Leftrightarrow \exists b((a, b) \in f \wedge (b, c) \in g) \Leftrightarrow \exists b(b = f(a) \wedge c = g(b)) \Leftrightarrow c = g(f(a))$.
   - The relation order is "first f, then g", while the function notation $g(f(a))$ reads from right to left.
2. **Injective.** Let $a, a' \in A$ with $g(f(a)) = g(f(a'))$.
   - g injective $\Rightarrow f(a) = f(a')$.
   - f injective $\Rightarrow a = a'$.

   Surjective. Let $c \in C$.
   - g surjective $\Rightarrow \exists b \in B : g(b) = c$.
   - f surjective $\Rightarrow \exists a \in A : f(a) = b$.
   - Hence $(g \circ f)(a) = g(f(a)) = g(b) = c$.

   In particular the composition of two bijections is a bijection (used in the proof of Lemma 3.15 (i)).
3. $\boldsymbol{f}$ **injective.** Let $f(a) = f(a')$.
   - Apply g: $g(f(a)) = g(f(a'))$, i.e. $(g \circ f)(a) = (g \circ f)(a')$.
   - $g \circ f$ injective $\Rightarrow a = a'$.

   $\boldsymbol{g}$ **surjective.** Let $c \in C$.
   - $g \circ f$ surjective $\Rightarrow \exists a \in A : g(f(a)) = c$.
   - So $b := f(a) \in B$ satisfies $g(b) = c$.
4. $A = C = \{0\}$, $B = \{0, 1\}$, $f(0) = 0$, $g(0) = g(1) = 0$.
   - $g \circ f = \mathrm{id}_A$, which is bijective.
   - f not surjective: $1 \notin f(A)$. g not injective: $g(0) = g(1)$, $0 \neq 1$.
   - So in 3. the roles of f and g cannot be swapped.

Equinumerosity

Problem 4 ⋆ ⋆

1. Let X be an infinite set, and let $a_0, a_1, a_2, \ldots \in X$ be pairwise distinct (an infinite sequence in X without repetitions). We show that removing a_0 does not change the "size" of X, i.e. $X \sim X \setminus \{a_0\}$.
   (a) Define a function $h : X \to X \setminus \{a_0\}$ that shifts the sequence by one position ($a_0 \mapsto a_1$, $a_1 \mapsto a_2$, $a_2 \mapsto a_3$, ...) and maps every other element of X to itself. Write h down as a case distinction.
   (b) Prove that h indeed maps into $X \setminus \{a_0\}$, that h is injective, and that h is surjective.
2. Use 1. to show $\mathbb{N} \sim \mathbb{N} \setminus \{0\}$.
3. Use 1. (twice) and Lemma 3.15 to show $[0, 1] \sim (0, 1)$.

Solution:

1. (a) Let $R = \{a_k \mid k \in \mathbb{N}\}$ be the set of sequence elements. Define $h : X \to X \setminus \{a_0\}$ by

$$h(x) = \begin{cases} a_{k+1} & \text{if } x = a_k \text{ for some } k \in \mathbb{N} \\ x & \text{if } x \notin R \end{cases}$$

   - Remark: yes, X must be infinite. This is already forced by the assumption: $a_0, a_1, a_2, \ldots$ are infinitely many distinct elements of X.

   (b)
   - h is a function: k is unique since the a_k are pairwise distinct. It maps into $X \setminus \{a_0\}$: $a_{k+1} \neq a_0$, and $x \notin R$ implies $x \neq a_0$.
   - Injective: let $h(x) = h(y)$. Both in R: $a_{k+1} = a_{m+1} \Rightarrow k = m \Rightarrow x = y$. Both not in R: $x = h(x) = h(y) = y$. Mixed ($x \in R$, $y \notin R$): $h(x) \in R$ but $h(y) = y \notin R$, impossible.
   - Surjective: let $y \in X \setminus \{a_0\}$. If $y = a_m$, then $m \geq 1$ and $y = h(a_{m-1})$. If $y \notin R$, then $y = h(y)$.
2. $X = \mathbb{N}$, $a_k = k$: $h(n) = n + 1$ is a bijection $\mathbb{N} \to \mathbb{N} \setminus \{0\}$.
3. - $X = [0, 1]$, $a_k = \frac{1}{k+1}$ (so $a_0 = 1$): $[0, 1] \sim [0, 1)$.
   - $X = [0, 1)$, $a_0 = 0$, $a_k = \frac{1}{k+1}$ for $k \geq 1$: $[0, 1) \sim (0, 1)$.
   - $\sim$ is transitive (Lemma 3.15 (i)), so $[0, 1] \sim (0, 1)$.
   - For finite X no such sequence exists, which matches §3.7.2: a finite set is never equinumerous to a proper subset.

Countable and Uncountable Sets

Problem 5 ⋆ ⋆

Decide whether each set is countable or uncountable. Justify with results from the script.

1. $\mathbb{Z} \times \mathbb{Q}$
2. $\mathcal{F} = \{S \subseteq \mathbb{N} \mid S \text{ is finite}\}$
3. $\mathcal{P}(\mathbb{N})$

Solution:

Tool: if B is uncountable and $B \preceq A$, then A is uncountable. (Otherwise $B \preceq A \preceq \mathbb{N}$, so $B \preceq \mathbb{N}$ by Lemma 3.15 (ii).)

1. **Countable.** $\mathbb{Z}$ is countable (Example 3.56), $\mathbb{Q}$ is countable (Corollary 3.21), so $\mathbb{Z} \times \mathbb{Q}$ is countable (Corollary 3.20).
2. **Countable.** Map $S = \{s_1, \ldots, s_n\}$ with $s_1 < \ldots < s_n$ to the tuple $(s_1, \ldots, s_n) \in \mathbb{N}^*$ (and $\varnothing$ to the empty sequence).
   - Injective: the sorted tuple lists exactly the elements of S, so different sets give different tuples.
   - $\mathbb{N}^*$ is countable (Theorem 3.22 (iii)), hence $\mathcal{F} \preceq \mathbb{N}^* \preceq \mathbb{N}$ (Lemma 3.15 (ii)).
3. **Uncountable.** $\chi : \{0, 1\}^\infty \to \mathcal{P}(\mathbb{N})$, $s \mapsto \{i \in \mathbb{N} \mid s_i = 1\}$, is a bijection (§3.7.4: a sequence specifies a subset of $\mathbb{N}$). So $\{0, 1\}^\infty \preceq \mathcal{P}(\mathbb{N})$, and $\{0, 1\}^\infty$ is uncountable (Theorem 3.23).

Problem 6 ⋆ ⋆

Let $\mathcal{I} = \{S \subseteq \mathbb{N} \mid S \text{ is infinite}\}$.

1. Prove that $\mathcal{I}$ is uncountable by giving an explicit injection $\varphi : \{0,1\}^\infty \to \mathcal{I}$. Prove that $\varphi(s) \in \mathcal{I}$ for all s and that φ is injective.
2. The map $\chi : s \mapsto \{i \in \mathbb{N} \mid s_i = 1\}$ from Problem 5 is injective, but it is not a function $\{0,1\}^\infty \to \mathcal{I}$. Why not?

Solution:

1. Define $\varphi(s) = \{2i \mid s_i = 1\} \cup \{2i+1 \mid i \in \mathbb{N}\}$: even numbers encode s, odd numbers make the set infinite.
   - $\varphi(s) \in \mathcal{I}$: $\varphi(s)$ contains all odd numbers, hence is infinite.
   - φ injective: let $s \neq s'$. Then $s_i \neq s'_i$ for some i, w.l.o.g. $s_i = 1$, $s'_i = 0$. So $2i \in \varphi(s)$, but $2i \notin \varphi(s')$ ($2i$ is even, and $2i = 2j$ only for $j = i$). Hence $\varphi(s) \neq \varphi(s')$.
   - So $\{0,1\}^\infty \preceq \mathcal{I}$. If $\mathcal{I}$ were countable, Lemma 3.15 (ii) would give $\{0,1\}^\infty \preceq \mathbb{N}$, contradicting Theorem 3.23.
2. A function into $\mathcal{I}$ must take only values in $\mathcal{I}$ (Def. 3.33). But e.g. $\chi(000\ldots) = \varnothing$ and $\chi(1000\ldots) = \{0\}$ are finite, so they are not in $\mathcal{I}$. The extra odd numbers in φ exactly repair this.

Problem 7 ⋆ ⋆ ⋆

1. Let A be countable and $f : A \to B$ surjective. Prove that B is countable.
   (Hint: use an injection $h : A \to \mathbb{N}$ to pick, for each $b \in B$, one element of $f^{-1}(\{b\})$.)
2. Prove or disprove: if A is uncountable and $f : A \to B$ is surjective, then B is uncountable.

Solution:

1. A countable means there is an injection $h : A \to \mathbb{N}$ (Def. 3.42).
   - For $b \in B$, $f^{-1}(\{b\}) \neq \varnothing$ (f surjective), so $h(f^{-1}(\{b\}))$ is a non-empty subset of $\mathbb{N}$. Let $g(b)$ be its least element (well-ordering principle, cf. proof of Theorem 3.17).
   - This defines a function $g : B \to \mathbb{N}$. By construction $g(b) = h(a)$ for some a with $f(a) = b$.
   - g injective: let $g(b) = g(b')$, i.e. $h(a) = h(a')$ with $f(a) = b$, $f(a') = b'$. h injective $\Rightarrow a = a' \Rightarrow b = f(a) = f(a') = b'$.
   - Hence $B \preceq \mathbb{N}$, i.e. B is countable.
2. **False.** $f : \{0,1\}^\infty \to \{0\}$, $s \mapsto 0$ is surjective, $\{0,1\}^\infty$ is uncountable (Theorem 3.23), but $\{0\}$ is finite. Surjections can only "shrink"; uncountability is inherited along injections, in the other direction (B uncountable, $B \preceq A \Rightarrow A$ uncountable).

Problem 8 ⋆ ⋆ ⋆

Let A be an arbitrary set.

1. Show $A \preceq \mathcal{P}(A)$.
2. Let $f : A \to \mathcal{P}(A)$ be any function and $D = \{a \in A \mid a \notin f(a)\}$. Prove that $D \notin f(A)$. Conclude $A \not\sim \mathcal{P}(A)$.
3. Deduce that $\mathcal{P}(\mathbb{N})$ is uncountable, without using Theorem 3.23.
4. Let $A = \mathbb{N}$ and identify each subset $S \subseteq \mathbb{N}$ with the sequence $s \in \{0,1\}^\infty$ where $s_i = 1 \Leftrightarrow i \in S$. Which object from the proof of Theorem 3.23 corresponds to D?

Solution:

1. $a \mapsto \{a\}$ is injective, since $\{a\} = \{a'\} \Rightarrow a = a'$ (Lemma 3.1).
2. Assume $D = f(d)$ for some $d \in A$. Case distinction:
   - $d \in D \Rightarrow d \notin f(d) = D$ (def. of D). Contradiction.
   - $d \notin D \Rightarrow d \notin f(d)$, so $d \in D$ (def. of D). Contradiction.

   So $D \notin f(A)$: no function $A \to \mathcal{P}(A)$ is surjective, in particular none is a bijection. Hence $A \nsim \mathcal{P}(A)$.
3. Assume $\mathcal{P}(\mathbb{N})$ countable, i.e. there is an injection $g : \mathcal{P}(\mathbb{N}) \to \mathbb{N}$. Define $F : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ by

$$F(n) = \begin{cases} S \text{ if } n = g(S) \text{ for some } S \\ \varnothing \text{ if } n \notin g(\mathcal{P}(\mathbb{N})) \end{cases}$$

   - Well-defined: S is unique since g is injective.
   - Surjective: every S satisfies $S = F(g(S))$.
   - This contradicts 2. Hence $\mathcal{P}(\mathbb{N})$ is uncountable.
4. Write $f(i) = \beta_{i,0}, \beta_{i,1}, \ldots$ as in Theorem 3.23. Then $\beta_{i,i} = 1 \Leftrightarrow i \in f(i)$, and

$$i \in D \Leftrightarrow i \notin f(i) \Leftrightarrow \beta_{i,i} = 0 \Leftrightarrow \overline{\beta_{i,i}} = 1.$$

   D is exactly the diagonal sequence α: "$d \in D \Leftrightarrow d \notin f(d)$" is "$\alpha$ differs from $f(d)$ in position d".
   - Iterating 1. and 2.: $\mathbb{N} \preceq \mathcal{P}(\mathbb{N}) \preceq \mathcal{P}(\mathcal{P}(\mathbb{N})) \preceq \ldots$, and no set in this chain is equinumerous to the next one.

Problem 9 ⋆ ⋆ (extra)

1. Let A, B be finite with $|A| = |B|$ and $f : A \to B$. Prove: f injective $\Leftrightarrow$ f surjective.
2. Show that both directions fail for $A = B = \mathbb{N}$.

Solution:

1. ($\Rightarrow$) f injective, so f is a bijection $A \to f(A)$ and $|f(A)| = |A| = |B|$. A subset $f(A) \subseteq B$ of a finite set with as many elements as B is B itself, so $f(A) = B$.

   ($\Leftarrow$) Contrapositive. If f is not injective, two elements of A share an image, so $|f(A)| \leq |A| - 1 < |B|$ and $f(A) \neq B$.
2. - $n \mapsto n + 1$ is injective, but 0 has no preimage.
   - $n \mapsto \lfloor n/2 \rfloor$ is surjective ($m = f(2m)$), but $f(0) = f(1)$.

Problem 10 ⋆ ⋆ ⋆ (extra)

Let $\mathcal{J}$ be a set of pairwise disjoint open intervals $(a, b) \subseteq \mathbb{R}$ with $a < b$.

1. Prove that $\mathcal{J}$ is countable. (You may use: between any two reals $a < b$ there is a rational number.)
2. Does 1. still hold for pairwise disjoint closed intervals $[a, b]$ with $a \leq b$?

Solution:

1. Let $q : \mathbb{Q} \to \mathbb{N}$ be an injection (Corollary 3.21).
   - For $J \in \mathcal{J}$, $J \cap \mathbb{Q} \neq \varnothing$ (hint), so $q(J \cap \mathbb{Q})$ is a non-empty subset of $\mathbb{N}$. Let $g(J)$ be its least element (well-ordering principle).
   - g injective: $g(J) = g(J')$ means $q(r) = q(r')$ with $r \in J$, $r' \in J'$. q injective $\Rightarrow r = r' \in J \cap J'$, and disjointness gives $J = J'$.
   - Hence $\mathcal{J} \preceq \mathbb{N}$.
2. **No.** $\mathcal{J} = \{[x, x] \mid x \in \mathbb{R}\} = \{\{x\} \mid x \in \mathbb{R}\}$ consists of pairwise disjoint closed intervals and is equinumerous to $\mathbb{R}$ ($x \mapsto \{x\}$ is a bijection), which is uncountable (§3.7.4). The proof of 1. breaks because $[x, x]$ contains no rational when x is irrational.